Unit 10 - Communication Technologies
Assignment 7: Upgrade a LAN Tasks 8&9

Introduction

Task description - *You are a junior consultant for T5 Alliance. Your new client, Competition Bikes Ltd, have decided that they would like to have wireless capability in their new network. You must adjust your design to include the following new requirements;*

- *Wireless coverage must extend to the whole site.*
- *The wireless network must be as secure as possible.*

Task 8

(U10P5) - Describe the basic principles of signal theory

Task description - *Description of the following principles of signal theory;*

- *Signalling methods (light, radio, electrical, microwave)*
- *Synchronous and Asynchronous transmission*
- *Error detection and error correction*
- *Bandwidth (limits and noise)*
- *Channel types (telephone lines, HF radio, microwave, satellite)*

Signalling methods (light, radio, electrical, microwave)

Data signalling methods, also known as Digital signal processing (DSP), is a group of various techniques which are used to send data over telecommunication, which work by sending a clear data signal. Essentially, data signals are signals transmitted by a device such as a computer or router.

Usually, the signals contain data in the form of binary codes, which consist of a pattern of 1's and 0's, which is understandable by computers better than the human brain due to the algorithmic function of binary.

The data signaling methods used are in waveforms, whether it is light, radio, microwave or even electrical. These waveforms are made up of two main characteristics, which is amplitude and frequency (wavelength). However, frequency is very important for data transmission, because it can mean the difference between transfer speed and range.

Frequency being the distance between each wave in the data transfer, which can be used to determine the strength of the signal, which is speed and range. Meaning that the higher the wavelength, the larger the range, giving it the ability to travel a longer distance to that with a

shorter wavelength. However, a shorter wavelength has much faster data transfer speed, but much shorter range and consumes much more power when transmitting data at similar speeds and distances achieved by lower frequencies.

These different data signaling methods are beneficial for a network, because there are many different scenarios which require different advantages. For instance, some networks might find distance over speed or speed over distance, it basically depends on the scenario.

Typically, there are multiple different types and methods that were and are utilised in different situations to harbour the best outcomes. For instance, fibre optic cables transmit data using wires that have light-emitting diodes at the end of them meaning that the data can be transmitted using light. This in turn is more expensive, however harbours greater data transmission speed and distance when compared to other methods, such as electrical.

Other examples would be radio and microwaves, which uses radio waves to send over information, usually from a wireless access point. Which is much cheaper than fibre optic, but in most cases a lot slower speeds and significantly lower distance when compared to both electrical and light, especially for microwaves.

Electrical is also another signalling method used for signalling methods, which can be found within copper wires such as Twisted pair. Which is much cheaper than both fibre optic and wireless data methods, but electrical signal method are falling behind in speed, as more modern wireless and fiber optic cables surpass the bandwidth speeds of electrical signal methods.

Overall, the different signalling methods, those being light, radio, electrical and microwave, use waveforms to send data in the form of binary. Each containing different methods of waveform, some of which have better speeds or distance than the other methods.

For instance, the wavelength for fiber optic are 850, 1300 and 1550 nm (nanometers), and the reason it varies is due to wavelength-division multiplexing (Which is done by changing the colour of the laser through the cable).

Meaning that it has an extremely short wavelength than can be altered, giving it a significant transfer speed which can be changed. However, this only has en effect on the range if it is not travelling in a straight line, because light can travel extreme distances in a straight line, meaning it can contain data for a long distance.

The wavelength for electrical waves, microwaves and radio waves are between 300 GHz to 3 kHz (radio waves), 300 MHz to 300 GHz (Microwave) and 1 MHz to 100 MHz (electrical waves within a twisted pair cable. Meaning that electrical waves has a much higher bandwidth speed than microwave and radio waves, while having much less range.

Microwaves have much faster bandwidth speed than radio waves but slower than electrical waves, and the range of microwaves is higher than electrical waves but smaller than radio waves. While radio waves have much higher range than both microwave and electrical waves, but a much slower bandwidth speed due to the wavelength.

Synchronous and Asynchronous transmission

Synchronous and asynchronous transmission is a method of determining whether a signal is or isn't synchronised between two hosts. What I mean by this is that both the hosts, typically two devices within the same network, both send and receive information. Meaning that synchronization is obligatory in order for both devices to communicate.

Synchronous transmission is when transmitted data being sent and received will be synchronised by a clocking signal. Meaning that 2 devices with the same clocking signal would be able to communicate with each other. For example, when a computer communicated with a network server.

Asynchronous transmission doesn't require a clocking signal as the data sent and received will not be synchronised. Typically, it is used when a device needs to communicate with another device and doesn't need it to communicate back. For example, when a computer connects to a printed

Essentially, both types of transmission is used for the correct communication between two hosts, which can be two computers within a network or a computer trying to access a printer, so communication between the two hosts are possible in the first place.

Overall, synchronous transmission is a method in which a sequence of data signals are sent accompanied by timing signals, this is to ensure that that the transmitter and the receiver are in sync together.

While asynchronous Transmission is the opposite to synchronous transmission since the data in this form of transmission is sent in spurts. Also, because a bit character must entered before each bit of is sent to inform the receiver when the transmission is finished.

Error detection and error correction

When sending data from host to receiver, it isn't guaranteed that it will always be correct or flawless, due to corrupted data or significant amounts of bandwidth noise. Which is why both error detection and error correction is essential, to make sure that information or data does not reach the host corrupted and that the data translates well.

Error detection is used to detect noise and other impairments within the data transmission, which ensures that the data that is being transmitted is reliable and reduces the chance of

incorrect frames and other forms of data transfer errors. This is used when data is being transmitted between two devices with the likely chance of there being errors within the data transfer, which would detect said errors and making it easier for the problem to be fixed.

Error correction is the process after the error detection, which is when the error is removed, therefore fixing the error that has been found using the error detection. This ensures that when the data is received by the other side, the data will be free of errors, therefore being a more reliable connection.

Overall, Error detection is a technique which is used to find an error found within the data, errors occur because of channel noise, which irregularities in the signal, which occurs during the sending of data.

Error correction occurs after the process of error detection has occurred. Usually after the packet of data has been sent, since it is in the trailer of the structure. Which reconstructs the data and therefore ensuring errorless data has been transmitted.

Bandwidth (limits and noise)

Bandwidth is a measurement of how much data can be transmitted through a single communication channel. Bandwidth is normally expressed in bits per second (bps). Modern networks now have the capability to transfer millions/billions of bits per second, meaning that gigabits of information can be transferred within a single second.

Bandwidth limitation is where a user is given a limit to the amount of data is allowed to be transmitted. What I mean by this is that it is the amount of data that is allowed to pass through at any given time. So, a higher bandwidth would mean that the data would be transmitted faster, due to the wider space for data traffic.

Bandwidth noise is what affects the quality of the data that is transmitted. What I mean by this is that the noise is the interference that has a negative impact of the reliability of the signals being sent over a network. Noise can occur due to many reasons, either internally or externally. Examples of the type of noise that might be experienced are other radio waves, lighting, nearby electrical cables.

Overall, bandwidth limitation, also known as a bandwidth cap are put into place by internet service providers, they limit the speed of data transfer on a broadband internet connection. So, he more the user pays the higher for their internet service such as fibre optic, the higher the limit of data transfer therefore allowing for a faster into speed.

While bandwidth noise is a unwanted electrical energy which decreases the quality of signals during transmission, it can cause errors in all types of data, such as text and audio. However, the effects of bandwidth noise can be aided by the user of the error detection and correction.

Channel types (telephone lines, HF radio, microwave, satellite)

Channel types are the transmission path that the signals, or data, choose to travel over. Essentially, it is the path between 2 nodes within a network. Typically, there are many different types of channels, those channels being: Telephone, HF radio, microwaves and satellite.

Telephone communication is the transmission of speech over a distance which occurs by electric signal conductors or by the use of radio signals.

However, telephone channel types are slower than the others, due to the old fashioned way of radio/electrical data transmission. Basically, telephone line communication occur over the use of an electrical or radio channel, which is a bit-stream of information sent from the transmitter to the receiver.

HF Radio waves is a radio frequency in which the waves are between 3-30 MHz, because this is a high frequency this therefore allows for the quick transmission of data over a network, such as mobile phones etc.

Microwaves are a form of electromagnetic radiation in which the length of the waves differ in size. An advantage of the use of microwaves is that no cables are used in the transmission, multiple channels are also available and data is transmitted at a high bandwidth.

Satellites can be used for communication, the definition of a satellite is an artificial device which orbits the earth for the purpose of communication. The advantage of communication via the use of satellites would be that they allow for a high bandwidth, are fairly cheap and also cover a large area due to their position.

Overall, Telephone is an old channel type that are slower than other channel types, due to the unreliable data transmission (bandwidth noise) and travelling via electrical waves over a long distance. Meaning that it sends a small amount of data when compared to other methods, over a small distance when also compared to other channel types(excluding microwaves).

Radio channels are used on high frequencies, used on common devices such as a radio or other devices used for human communication without physical contact. Meaning that it sends a small amount of data at a consistent rate, but over a large distance.

Microwaves are used by devices such as computers and other mobile devices which has a WiFi connection. Meaning that it can send large amount of data but over a limited distance when compared to other channel types.

Satellite is a channel type which transmits data at a high speed, which is often used on common devices such as televisions. Meaning that it can send large amounts data when compared to

other channel types, apart from microwaves, over a large distance when compared to other channel types.

Conclusion

In conclusion, the principles of signal theory is important for a network, because it essentially distinguishes the different types of signals and transmission methods which can be used within a network.

This is so the correct signal can be used for the specific scenario within a network. For instance, the need for connectivity distance over connectivity speed can prove to be more important in some situations, while the same can be said for the opposite (Speed over distance).

Also, the recognition of errors within a network and how to overcome said errors is vital to understand. This is so the network can carry on its intended purpose without being plagued with errors such as bandwidth noise and limit and how to repair said errors using error detection and error correction.

Overall, understanding signal theory helps understand what type of transfer method to use, why use said method and how. This is so the network works in the most efficient manner possible, fitting to the scenario that the network is currently in.

Task 9

(U10P4) - Describe data elements and explain their importance

Task description - *Description of the following data elements:*

- *Checksum (Cyclic Redundancy Check)*
- *Encapsulation (segments, packets, frames)*
- *Addresses (physical and logical)*
- *Sequence numbers*

Checksum (Cyclic Redundancy Check)

CRC (Cyclic redundancy Check), or Checksum, is an error-detecting code which is used both on storage devices such as hard drives or in computer networks. Essentially, CRC is a simple error detection system in which each message which is transmitted, accompanied by a value.

Checksum is important because it can be used to count the number of bits which is being transmitted in a single transmission of the data. This is important because it can be used by the receiver to see if the same amount of bits has been received that was sent, therefore ensuring that the data is not corrupt.

This is checked when data enters the system, which is given a short check value. This enables checks to occur quickly, that would alter if a change to the data was to occur. This is important because even small changes to data in a network can have a big effect such as stopping a network working compactly.

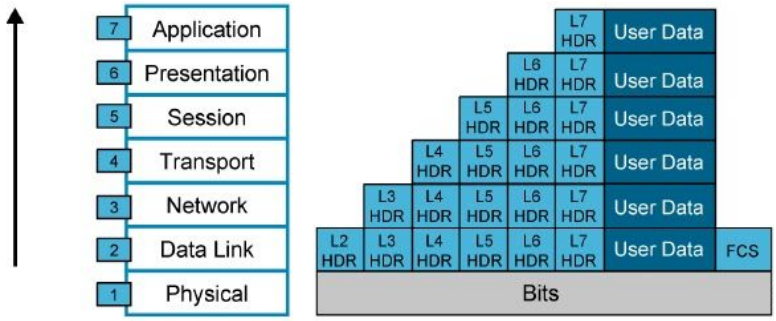
Overall, Checksums are an example of data elements that are used in the process of error detection. So, when a process has been executed correctly, the digits will add up to a number that indicates that the process has been executed correctly. If this is not the case, then we know that an error has occurred, meaning that the error can be dealt with and resolved with ease.

Basically, this gives the ability to carry out check runs to detect any accidental changes, which helps pinpoint the location of the error. Also, checksums can be used to ensure that data has not been corrupted, preventing from any error occurring in the first place.

Encapsulation (Segments, packets, frames)

Encapsulation is defined as the packing of data/functions into a signal component. Examples of this include frames, packets and addresses. A data element, which is either a frame, packet or an address, is essentially a unit of data that is used when data is transmitted over a network.

An example of a data element is a data packet which is a essentially a chunk of a much larger piece of data which was cut into smaller chunks in order for it to be transmitted.



HDR = Header

All communication devices will transmit data elements, electronically, over a network as all devices pertain to either the TCP/IP model or the OSI model, as seen in the image above. In the TCP model all data is broken down into packets, however in the OSI model the data elements are known by different names depending on the layer of transmission.

On the top three host layers they are merely ‘data’. However, when the data flows to the transport layer it is known as a segment. It is only when data moves from the host layers to the media layers, most notably the network layer, that the segment becomes a packet. As it is

passed down to the data link layer and then physical layer it becomes a series of frames and then bits.

A packet is a unit of data which is carried across a network, they are transferred in this manner because this the bandwidth of communication can be better and stronger between users when it comes to network communication.

A pack of data consists of kind different types of data, the first of these is control information which provides information to the network in order for the packet to be deliver to the right location, the second part is the user data which is the information which the user wants to transmit an example of this would be a word document etc.

A frame in computing is defined as a transmission unit which is used in a computer network. A frame consists of bits of information which is transferred over a computer network. An example of where frames are used on computer networking would be on the Open System Interconnection model (OSI model).

Within the model a frame is vital into the second layer because layer 2 is known as the data-link layer. The role of this layer is to create links across a network, this is done by gathering information and transferring data in the forms of frames.

Overall, encapsulation is basically the process of splitting data up into smaller segments of data, to then be re-encapsulated at the destination, recreating the data created from the host. This is done so the data can be transferred over different layers of the OSI model, which is vital in telecommunication within a network.

The reason that it is vital is because there are many types of devices within the network, some more “intelligent” than others. What I mean by this is that they can only handle and send certain amounts of information over a network, hence the OSI model. Splitting the data up into smaller bits, gives the ability for devices operating at the lower levels of the OSI model to send the data over the network.

Addresses (Physical and logical)

A network address is a an address and is used an identifier on a telecommunications network, it allows certain web pages to be searched for. In order for an address to be effective, the network address must be unique within the network. In a network, there are two main types of network addresses, which are MAC addresses and IP addresses.

MAC addresses are a globally unique code made up of a 48-bit hexadecimal number coded to the device, which is required by the manufacturer of the device. Hosts within a network use a destination MAC address to sends a frame over a switched network, so it can be used to build or address said frame.

IP addresses are either made up of 32-bit binary code (IPv4) or 128-bit hexadecimal code (IPv6), which is given to a computer by the router.

The reason that different IP addresses are given to specific computers is due the networks requiring to send data or receive data from the computer. So it needs to knows which computer to send it too, doing so through the IP address. This makes it a lot easier for the devices within networks to communicate with each other.

Addresses can also be used to identify and search for web pages, for instance 'http://www.google.com/' is the web address for google. However, it is not the actual identity of the address, it is converted to text because it is easier for humans to remember words instead of numbers. The actual address of "google.com" is either 8.8.8.8 or 8.8.4.4 in IPv4 configuration, and 2001:4860:4860::8888; 2001:4860:4860::8844 in IPv6 configuration.

Overall, addresses are essential for telecommunications, especially at the layer 2 and layer 3 of the OSI model. The reason being that it gives each device within the network it's own identity, meaning that the data can be sent accurately to the correct destination in an efficient method. Without the data either being sent to every device, which would be frustrating for users, or the data being lost in general because it didn't reach its host.

Sequence numbers

Sequence numbers, also known as transmission sequence numbers are numbers that are given to data fragments via the SCT Protocol. Sequence numbers are numbers that are in a certain order, which are referred to as a string.

These numbers let the clients know which order the data fragments have to be assembled in before they can be processed, meaning that it ensures that the data arrives at the receiver in the network. Essentially, the numbers sort out error correction and identification for data transfer.

An example of a sequence number protocol which is used to increase reliability within a network, is the TCP protocol. This protocol controls the packets of data, by giving packets of data sequences of numbers before being transmitted over the network, basically acting as an identification number.

With these given numbers, the host at the receiving end will examine each number on the packets to see if all the packets were delivered correctly. If it were on the chance where a packet or more were missing, the receiving host would send an acknowledgement packet back to the host who sent said packets, which would say there were some missing packets so they could be resent.

An acknowledgement packet can also be sent to confirm a successful transfer if all the packets arrived correctly. Also, the sequence of numbers that are given to the packets can be used to reorder the packets into the order they were in before being transmitted, meaning that it is less likely for data to be corrupted.

Overall, sequence numbers are 32 number long sequence which is used by stream control transmission protocol, the numbers are attached to packets containing user data because it allows the packet to acknowledge the endpoint of the route and send receipt.

The reason that sequence numbers are used in computer networks is to make sure that data actually arrives at the receiving end of the network from the host, by using error correction and identification.

Conclusion

In conclusion, it is important to understand the data elements and their importance because it gives knowledge on how to send data over a network efficiently and reliably. Also to understand the different precautions to take when something goes wrong.

The reason being because it essentially gives the knowledge required to know how the data is being transferred and what to do when there are errors. For instance, using Checksum or the use of sequence numbers to check for errors within a network, such as lost data packets and sorting out packets in an efficient manner.

Also, the recognition of what type of data to send over the network, using knowledge from encapsulation and what layer on the OSI model the devices within the networks are on. This is so the network can send data correctly, without causing significant errors or losing a lot of important data due to overloading a device with information, which may also cause the device to malfunction.

Overall, understanding data elements makes the network carry out its purpose without being interfered with unfortunate events such as data loss, malfunctions or other possible errors that could occur within a network.